

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (2 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1** **$\frac{1}{2}$ point**

Why should the gas pressure in the vacuum tube of a Franck-Hertz experiment be low?

Question 2 **$\frac{1}{2}$ point**

Draw an atomic model of a ground-state argon atom.

Question 3 **1 point**

Calculate the binding energy of the outermost electron in the Ar atom. According to Bohr's model, the energy of an electron in the n^{th} orbital of an atom of atomic number Z is

$$E = -13.6 \frac{Z^2}{n^2} \text{ eV}.$$